

Computability: Exercise 3

Due: January 24, 2024

Question 1

Consider the following two DFAs $\mathcal{A}$ and $\mathcal{B}$.

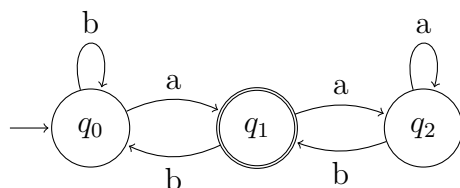


Figure 1: $\mathcal{A}$

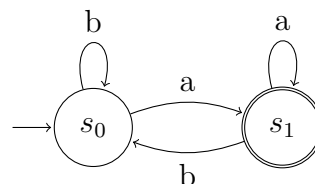
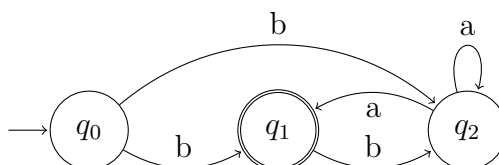


Figure 2: $\mathcal{B}$

- Draw the product automaton for $L(\mathcal{A}) \cup L(\mathcal{B})$. There is no need to draw unreachable states.
- Draw an NFA for $L(\mathcal{A}) \cup L(\mathcal{B})$, and then determinize it.
- Prove that for any two automata, the product automaton is precisely the DFA obtained by determinizing the NFA for their union.

Question 2

Consider the following NFA $\mathcal{A}$ over $\Sigma = a, b$:



- Run the word $w = baababbb$ over $\mathcal{A}$ using subset construction simulation (as we saw in the recitation.), and determine whether this word is accepted or not. In your answer, show the states in each iteration of the run.
- Determinize $\mathcal{A}$ using the subset construction algorithm. There is no need to draw unreachable states. When you describe the equivalent DFA, indicate for each state the subset to which it corresponds.

Question 3

Let $\Sigma = \{a, b\}$, and let $L = \{w \in \Sigma^* \mid \#_a(w) = 1 \pmod{3}\}$.

- Find all Myhill-Nerode equivalence classes for L , find a representative for each class, and show a separating suffix for every pair of classes.
- Show an automaton deciding L that has more states reachable from the initial state than the number of Myhill-Nerode classes.

Question 4

Prove that the following languages are not regular.

- a. $L = \{1^n : n \geq 0\}$ over $\Sigma = \{1\}$.
- b. $L = \{a^i b^j c^k : i, j, k \geq 0, k = i \cdot j\}$ over $\Sigma = \{a, b, c\}$.
- c. Let $\Sigma = \{a, b, c\}$, and let

$$L = (\{a\} \cdot \{b, c\})^* \cup \{w \in \Sigma^* : \#_b(w) = \#_c(w)\}.$$

Hint: Take some time to understand the language L . Note that, for example, $aa, abab, aabc, baca \in L$ and $aab, baba, aabb \notin L$.

Question 5

In the proof of the Myhill-Nerode theorem in class, we skipped the proof that $\delta^*([e]_L, w) = [w]_L$ for every $w \in \Sigma^*$. Write down the full proof of this claim.

Question 6

Let $k \in \mathbb{N}$ (k is static), and let $L_k = \{1^i \cdot y \mid y \in \{a, b\}^k, 1 \leq i \leq k, \text{ and } y_i = a\}$

- a. Show DFA's for the languages L_k whose number of states is bounded from above by a polynomial in k .
- b. Prove that in any DFA that decides L_k^{reverse} , the number of states must be exponential in k . (recall that $L_k^{\text{reverse}} = \{w_{\text{rev}} \mid w \in L_k\}$, where for $w = \sigma_1 \dots \sigma_n$ we define $w_{\text{rev}} = \sigma_n \dots \sigma_1$).

Question 7

Prove that for the language $L = \{1^{2^n} : n \in \mathbb{N} \cup \{0\}\}$ over the unary alphabet $\Sigma = \{1\}$, no 2 words are in the same Myhill-Nerode equivalence class with regards to L .